

Tessera

The credit layer for agentic commerce.

2.0 · Encode Club Solana Hackathon 2026 · Best Use of x402 on Solana

Edison Liu · Computer Engineering, ZJU-UIUC Institute

X402, THE PAYMENT PROTOCOL FROM COINBASE

AI agents now *pay each other.*

≈200M

x402 payments so far

7¢

average payment on one recent day

Solana Foundation, 5 Aug 2026: “roughly 200 million transactions … Most transactions are under 50 cents.” x402scan.com, 24 h to 5 Oct 2026: 118,341 payments, $8.01K.

PAIN 1 · WHO IS ON THE OTHER END?

One person, unlimited agents. *No ID.*

And an x402 payment is final once it settles. No chargeback.

Behaviour | Amounts settled | Ratings

Fix 1: a credit layer that labels each anonymous agent by how it behaves.

Creating a Solana wallet needs no identity and costs nothing beyond fees. An x402 exact payment is a signed token transfer with no chargeback.

PAIN 2 · THREE ATTACKS WE HAD TO PRICE

Sybil

One person, many wallets.

Friendly fraud

Got the goods, claims they never came.

Reputation farming

Trade with yourself until you look trusted, then cash in on strangers.

Fix 2: escrow, *until the wallet has earned trust.*

Faking a record now costs real fees, and months you can't skip.

Mainnet targets: a new wallet counts 10%, each counterparty is capped, Trusted needs 30 active days; a farming ring costs hundreds to a few thousand dollars and one to three months. Priced, not impossible (docs/SECURITY.md).

FIX 2 · ESCROW THAT KNOWS WHO IT IS DEALING WITH

Escrow that knows *who it is dealing with.*

Every wallet has an on-chain score, built from settled orders, ratings and behaviour.

○○○○	New	3 d hold
●○○○	Building	24 h hold
●●●○	Established	60 min hold
●●●●	Trusted	instant

Mainnet target holds. On devnet the same four tiers hold for 2 min, 45 s, 10 s and 0.

NOTHING NEW TO ADOPT

One field changes: *where payTo points.*

```
HTTP/1.1 402 Payment Required
{
  "scheme":  "exact",
  "network": "solana:EtWTRABZ…",     // devnet
  "amount":  "200000",
  "payTo":   "JE6F5J7i…WdUW"     // an escrow account, not the merchant
}
```

A2A to ask x402 to pay a facilitator settles a Solana program holds

A real order from this project on devnet. The x402 exact scheme derives the token account from payTo, so an escrow account is a valid payTo.

REVIEWS THAT PAY

Paid to review. *Paid more for being right.*

Three buyers reviewed Glib. Then Glib took an order and never delivered.

Scout ★★★★★	0× · praised it, then it failed	$0.00
Nova ★★★★☆	0× · praised it, then it failed	$0.00
Orbit ★★☆☆☆	1.5× · warned, then it failed	$0.045

Paid on chain: 0.169 USDC of 1.25 USDC in fees, to 6 reviews

Same pay for 1★ or 5★

A quarter of the fee rate on the money behind the review. Stars never change it.

Judged on what happened next

An honest warning is paid 1.5×. Praising a wallet that then failed is paid nothing.

Faking it loses money

At most 75% of an order's fee comes back, so farming with your own wallets never pays.

Recorded 2026-10-10: real purchases, reviews, a missed delivery and the rebate payout, on a local validator running the real program. Reproduce: npm run build:program && npm run demo:rewards

PAIN 3 · TOP UP BEFORE EVERY JOB, COUNT THE TOKENS · FIX 3, NEW IN 2.0: PAY LATER

Buy now, pay later. *Still through escrow.*

The merchant quotes opens an escrow order; its 402 names it as payTo	0.20
The pool pays the order buy_now: into the escrow vault, never to the merchant	0.20
The escrow pays the merchant after delivery and the hold, minus its 1% fee	0.198
The buyer repays the pool the price plus 2%, by the due date	0.204

A 0.20 USDC purchase on credit. Credit never skips buyer protection: refund and dispute work as for any order.

Credit from the record

Only a wallet with a Tessera record gets a limit: tier limit × score ÷ 1000. A New wallet gets nothing.

Pay later

The price plus 2% by the due date, no interest. Paying on time raises the limit, up to double.

Or default, in public

Anyone can mark a late loan. The line freezes at zero for good.

New in 2.0. buy_now (programs/tessera-paylater) pays the order's vault and has the escrow confirm it in the same instruction; the escrow program is unchanged. Fees: the escrow's 1%, the pool's 2%. Tested by npm run test:paylater.

X402 · A SECOND WAY TO PAY, IN THE SAME 402

The 402 also offers *tessera-paylater.*

```
HTTP/1.1 402 Payment Required
"accepts": [
  { "scheme": "exact",            "payTo": "<order>", "amount": "200000" },   // any x402 client
  { "scheme": "tessera-paylater", "payTo": "<order>", "amount": "200000",
    "extra": { "feePayer": "<merchant>", "buyerLimit": "602000",
               "fee": "4000", "termSecs": 600, "eligible": true } }
]
```

1 · The buyer signs

One buy_now for that order, with the merchant as fee payer.

2 · The merchant settles it

It checks the transaction, simulates it, signs as fee payer and sends it.

3 · Then it delivers

The pool has paid the vault and the escrow has confirmed it, in one instruction.

Standard x402 clients are unaffected: the exact entries come first and pay as before. Shape from apps/agents/src/paylater.ts; buyerLimit is 2 USDC × 301 ÷ 1000, a Building buyer at score 301 on the devnet pool. Tested over HTTP, the CLI and A2A: npm run test:paylater-402.

LIVE ON SOLANA DEVNET

Two programs and a pool, *on devnet.*

Escrow program	TessSeP5QV5Bxpvm73iEefdsTjgtTEokKSx7jqFn1CQ	version 1, unchanged
Pay-later program	LXNPYNYcjv1G3HEJzurFrnj11NGnriy5xxqKSeTQU7D	new in 2.0 · upgraded 27 Oct
Pool	GNf8LpVeJ6ayrVMqABpU1nAHW9ZX7upC2aGqcumWF5Qq	funded with 8 devnet USDC
Pool vault	DK9angk2VJ43sUaUgJB9wpvDmjh4zroVj8zC2UYTXsgV	the pool's USDC
USDC (devnet)	4zMMC9srt5Ri5X14GAgXhaHii3GnPAEERYPJgZJDncDU	Circle's devnet mint

Pool rules: limits 0 / 2 / 5 / 10 USDC by tier · fee 2% · term 600 s + 60 s grace · merchants at Building or above

Read on chain on 27 Oct 2026 (getAccountInfo). Open any of them at explorer.solana.com/address/<address>?cluster=devnet. Time is compressed on devnet, so a whole loan can be watched in about ten minutes.

HONEST LIMITS

What it is not, *yet.*

- **Devnet test money.** Circle's devnet USDC, and the traffic is this project's own agents.
- **One pool authority key.** One lender, one pool: that key alone sets the rules and withdraws.
- **Unaudited.** Both programs have had internal reviews only.
- **A refund goes to the buyer, and the loan stays owed.** The buyer repays out of the refund, or defaults.
- **A default freezes the line, not the score.** The escrow program does not know about loans.
- **Devnet's escrow is an older build.** It is the 4 October build and its keys are lost, so the audit fixes run only on a local validator.

docs/PAYLATER.md, "Limits", and README, "What is real and what is not".

Micropayments were the
easy part.

Every payment *matters.*

agentic-commerce-two-theta.vercel.app

Live on Solana devnet · github.com/edison9733/x402_bnpl